Academic Year 2026-27
FY MCA- SEM I
IT12 Data Structure and Algorithms
Practice Assignment1

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Code:

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

class LinkedList:
    def __init__(self):
        self.head = None

    def create(self):
        n = int(input("Enter number of Nodes: "))
        for i in range(n):
            data = int(input(f"Enter values of {i}th Node: "))
            self.insert(data)

    def insert(self, data):
        node = Node(data)

        if self.head is None:
            self.head = node
        else:
            temp = self.head

            while temp.next:
                temp = temp.next

            temp.next = node

    def insert_idx(self, data, idx):
        node = Node(data)

        if idx < 0:
            print("Enter valid index")
            return
```

```
        if idx == 0:
            node.next = self.head
            self.head = node
            return

        temp = self.head

        for i in range(idx-1): # [0, 1, 2, 3]
            if temp is None:
                print("idx is more than size of linkedlist")
                return
            temp = temp.next

        if temp is None:
            print("Invalid index")
            return

        node.next = temp.next
        temp.next = node


    def traverse(self):
        temp = self.head

        while temp:
            print(temp.data, end=" -> " )
            temp = temp.next

        print("None")

    def middle(self):
        slow = self.head
        fast = self.head

        while fast and fast.next:
            fast = fast.next.next
            slow = slow.next

        if slow:
            print(slow.data)
        else:
            print("list is empty")

    def delete(self, idx):
        if self.head is None:
            print("list is empty")
```

```
        return

    if idx < 0:
        print("Enter valid idx")
        return

    if idx == 0:
        self.head = self.head.next
        return

    temp = self.head

    for i in range(idx - 1):
        if temp.next is None:
            print("Invalid index : ")
            return
        temp = temp.next

    if temp.next is None:
        print("Invalid Position")
        return

    temp.next = temp.next.next


def reverse(self):
    prev = None
    current = self.head

    while current:
        nextNode = current.next
        current.next = prev
        prev = current
        current = nextNode

    self.head = prev

def pair_sum(self):
    temp = self.head

    while temp and temp.next:
        print(temp.data, " -> ", temp.next.data, "\nsum : ", temp.data + temp.next.data)
        temp = temp.next
```

ACCREDITED WITH GRADE A+ NAAC [CGPA-3.32] in Cycle 1

```
def main():
    ll = LinkedList()

    print("####### 1. Create a LinkedList #######")
    ll.create()

    print("\n###### 2. Traverse a LinkedList #######")
    ll.traverse()

    print("\n######## 3. Insert at random position ########")
    idx = int(input("Enter position: "))
    data = int(input("Enter data: "))
    ll.insert_idx(data, idx)
    ll.traverse()

    print("\n######## 4. Middle of List ########")
    ll.middle()

    print("\n######## 5. Delete Node ########")
    idx = int(input("Enter position: "))
    ll.delete(idx)
    ll.traverse()

    print("\n######## 6. Reverse LinkedList ########")
    ll.reverse()
    ll.traverse()

    print("\n######## 7. Sum of every two consecutive nodes ########")
    ll.pair_sum()

    print("\n######## End of program ########")

main()
```

Output :

```
PS E:\Pranav\IMCC\Assignments\DSA> py .\10.10.2026.py
####### 1. Create a LinkedList #######
Enter number of Nodes: 4
Enter values of 0th Node: 0
Enter values of 1th Node: 1
Enter values of 2th Node: 2
Enter values of 3th Node: 3

###### 2. Traverse a LinkedList #######
0 -> 1 -> 2 -> 3 -> None

####### 3. Insert at random position #######
Enter position: 4
Enter data: 4
0 -> 1 -> 2 -> 3 -> 4 -> None

####### 4. Middle of List #######
2

####### 5. Delete Node #######
Enter position: 3
0 -> 1 -> 2 -> 4 -> None

####### 6. Reverse LinkedList #######
4 -> 2 -> 1 -> 0 -> None

####### 7. Sum of every two consecutive nodes #######
4  ->  2
sum :  6
2  ->  1
sum :  3
1  ->  0
sum :  1

####### End of program #######
PS E:\Pranav\IMCC\Assignments\DSA>
```